



HUMAN TRANSIT

How Clearer Thinking

about Public Transit
Can Enrich Our
Communities
and Our Lives

Jarrett Walker

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Communities and Our Lives

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INTRODUCTION

Suddenly, public transit is a critical issue.

For decades, the private car has seemed the ideal tool for free and spontaneous travel, and in rural areas and many small towns, that will continue to be true. But all over the world, people are moving into cities, and great cities just don't have room for everyone's car. Meanwhile, the converging threats of climate change and the end of cheap oil are forcing a new assessment of how cities work. Public transit—the most efficient means for large numbers of people to move freely within cities—is an essential tool in that effort. Today, even Los Angeles, a city that is world famous for its extreme dependence on cars, is scrambling to grow its transit system as fast as it can manage.¹

The frustrations of urban transportation have fed a growing public interest in public transit in many countries. But when well-intentioned people look at the public transit around them, many conclude that it does not make sense for them to use it. There are many good reasons not to use transit. It may not go where you need it to go, or at the time you need to go. Perhaps you can get there three times faster by driving, or at half the cost. You may know from experience that you can't trust your service to come on time. Your transit agency's vehicles and staff may fail to meet the most basic standards for civility, safety, and comfort. Your transit network may be too confusing, requiring you to wade through too much detail to figure out whether the service is useful to you.

Even if you decide that you can't use it yourself, you may support transit in hopes that *someone* will use it, because there is obviously not enough

room in your city for everyone to drive, or enough space to build more freeways and to widen streets. You may also understand the many kinds of long-term environmental harm associated with too many cars and too much pavement. You may even fear that driving your car while wanting fewer cars in your city is some sort of hypocrisy on your part. (Often it's not. You may live and travel in an area where no reasonably efficient transit can ever provide you with useful service.)

These common frustrations explain why, in many urban regions, support for public transit is wide but shallow. Voters typically support transit in general, but most don't know how to decide whether a transit proposal is good or bad. They may want better transit for themselves or their communities but have no idea how to make it happen.

What is quality public transit? Who can realistically expect to be served by it? What kinds of quality matter? How do we recognize and nurture them? What are the goals we want transit to achieve, and how do we navigate tough spots where these goals conflict?

Debates about transit proposals commonly lose track of these questions. Too often, we defer to a small group of intensely interested people (such as developers, activists, neighborhood groups, labor unions, and purveyors of transit technology) because the debate seems too technical for most of us to follow. As a result, we sometimes end up with transit investments that don't really do what we expected, or that have side effects that should have been foreseen.

Transit debates also suffer from the fact that today, in most of our cities, most of our decision makers are motorists. No matter how much you support transit, driving a car every day can shape your thinking in powerful, subconscious ways. For example, in most debates about proposed rapid transit lines, the *speed* of the proposed service gets more political attention than how *frequently* it runs, even though frequency, which determines waiting time, often matters more than speed in determining how long your trip will take. Your commuter train system will advertise that it can whisk you into the city in 39 minutes, but if the train comes only once every 2 hours and you've just missed one, *your* travel time will be 159 minutes, so it may be faster to drive, or even walk.

I can explain the concept of frequency to a group of well-intentioned motorists, and they may understand why it's important. But what they know is the experience of driving, where speed matters and frequency doesn't. So

when they make a decision about a transit project, they are likely to give frequency too little weight. The result can be services that are very fast but don't come when we need them, or that require too much time to connect from one service to another.

The unconscious assumptions of motorists are just one example of how people try to think about transit as though it were something else. Everyone tries to translate a question into terms that they understand. Economists may talk about transit in terms of profitability, as though that were its goal. Social service advocates think of it as a tool for meeting the needs of the disadvantaged. Architects and urban designers care about how it feels to move through a city, so they often focus on the aesthetics of the transit vehicle and infrastructure. Urban redevelopment advocates categorize services according to how well they stimulate development. None of these perspectives is wrong; transit can serve all of these interests and more. But to achieve that broad level of service, these points of view have to be brought together into a clearer conversation.

To aid that conversation, this book aims to give you a grasp of how transit works as an urban mobility tool and how it fits into the larger challenge of urban transportation. This is not a course designed to make you a qualified transit planner, though some professionals will benefit from it. My goal is simply to give you the confidence to form and advocate clear opinions about what kind of transit you want and how that can help create the kind of city you want.

WHO IS THE PUBLIC? WHO IS AN EXPERT?

When our elected leaders make decisions about transit, they face a noisy mix of competing interests. A senior citizen has trouble walking to a bus stop, so wants the stops placed closer together. Others want the bus stops farther apart, so that the buses run faster. A merchant wants the bus to deviate into his shopping center, to bring customers. Another merchant wants the bus out of his shopping center, because it's bringing "undesirable" people. Suburb A wants a proposed rail transit line to go underground through their community, to preserve their ambience. Farther out on the proposed line, Suburb B wants the whole line elevated, so that the line is cheap enough to get all the way to Suburb B in its first phase.

Transit professionals are not always in a position to clarify the debate. Some of them lack a sufficiently broad view of their product or have been trained to think only about one aspect of it. Many more have the understanding but lack the confidence. Even worse, professionals who work inside transit agencies often find their time consumed by daily crises and controversies and simply don't have time to take a wider view.

For a variety of reasons, transit planning has not evolved as a credentialed discipline—like law, engineering, or architecture—where everyone has to pass a particular course of studies before they can be licensed to work. Transit agencies often value “real-world” experience more than education, but there's no agreement in the industry about which world is real or which experience is useful. In my work as a transit planner, I have met many managers of transit planning who had been hired directly from other fields, with expertise in planning, perhaps, but not in transit. I have met bus drivers who had been transformed into managers, knowing nothing of the craft of management. Some of these people learn fast, adapt, and thrive, but others feel defensive and turn into obstacles.

The openness of the transit planning field—the ease with which it can be entered from related areas—has many advantages. The last thing we need is another closed and revered priesthood enforcing a uniform dogma, like the generation of highway engineers who designed America's interstate system. The principles of transit planning are simple enough that nobody needs a graduate degree to understand them. Anyone who is willing to keep learning should be welcomed into the transit professions.

On the other hand, there really are some facts about how transit works, and they are not all intuitively obvious. In fact, some of them will seem intuitively wrong until you stop to think about them. Most of these facts arise from math, geometry, and occasionally physics, so they're true everywhere, of every technology, and in every culture. If you're going to form intelligent opinions about transit, so that you advocate projects that actually serve your goals, you'll need to understand these basics.

These underlying facts of transit force us to make hard decisions, but they also open up possibilities. My aim is to help you see the unavoidable hard choices and to form your own view on them, but also to help you feel optimistic about the range of things transit can do and how a smart use of transit can improve your community. Like any box of tools, transit can do a lot of useful things, but only if you know each tool's purpose. Much of the

noisy confusion in transit debates is the sound of people using a hammer to turn a screw, or a screwdriver to pound nails.

LISTEN TO YOUR PLUMBER: VALUES VERSUS EXPERTISE

A core idea of this book is that we will have clearer conversations, and make better decisions, if we distinguish carefully between *values* and *expertise* and understand their interplay in our transit debates.

Values are statements about your community's ideals, goals, and priorities. They answer questions like these:

- *What is transit's purpose?* How should we measure the results of our transit system? Ridership? Emissions? Complaints?
- *What counts as adequate and useful transportation?* What, for example, is the minimum level of quality that transit should be aiming for?
- *What kind of city do you want?* Transit, like all transportation infrastructure, can have big impacts on the form, feel, and functioning of your city, so it's important to understand those impacts in advance.

Experts like me can clarify these questions but shouldn't be answering them for you. *My job in this book is not to make you share my values but to give you the tools to clarify and advocate yours.* You, and your community, get to choose "what" you want and "why." An expert's job is to help with "how." It's a crucial distinction, one that often gets lost in transit debates.

But here's the catch: *the expert gets to ask you questions that clarify what you want.* Say you hire a man to fix your plumbing. He goes to work, but soon he encounters a point where he could do one thing or another and it has to be your decision. He says: "I can fix it up for now for \$50, and it'll work for a year or so. Or, if I replace the whole whatsit assembly and connect it with a new doohickey, it'll be just like new, but that would be about \$700 and it would take a week for the part to get here from Malaysia."

The plumber's question reveals that values ("what") and expertise ("how") have to interact more than once. A transit planner working for your community is like a plumber: he's there to implement your values, not his. But you can't just tell the expert what you want and leave the room. When a leader or manager does this ("Just do it this way—I don't care how

you do it!”), he’s likely to be unhappy with the results. The values and the expertise must engage in a conversation.

Fortunately, as with plumbing, the questions that transit experts will have about your values are predictable. The same kinds of questions come up over and over. For this reason, the best way to form a resilient and credible opinion about transit is to think carefully about these typical “plumber’s questions” and to discuss them within your community. This book is designed to help you do that.

By definition, these questions are hard, because they’re about choosing between different things that you like. Your plumber is asking you if you’d like to save money now or have a more permanent fix. “Both” is not a useful answer. A transit expert helping your community has to ask many similar questions. Here are some of the big ones that we’ll explore:

- Is transit mostly about serving a peak-period or “rush hour” commute pattern, or is its top priority to provide a consistent service all day? (chapter 6)
- Would you rather have a direct but infrequent service or a more frequent service that requires a connection? (chapter 12)
- Is the goal of your transit system to carry as many people as possible? Or to serve disadvantaged persons who really need the service? Or something else? Or is it a balance of these, and if so, where do you strike that balance? (chapter 10)

These questions arise, unavoidably, from the underlying geometry of transit. This book explores each question in detail. Many people are trying to make transit do things that are geometrically impossible, so it’s important to start by exploring how transit works in these terms before going on to the question of technology.

TECHNOLOGY: TOOL OR GOAL?

When someone asks me what I do, and I say I’m a transit planner, their next question is almost always about technology. They ask my opinion

about a rail transit proposal that's currently in the news, or ask me what I think about light rail, or monorails, or jitneys. They assume, like many journalists, that the *choice of technology* is the most important transit planning decision.

Technology choices do matter, but the fundamental geometry of transit is exactly the same for buses, trains, and ferries. If you jump too quickly to the technology choice question but get the geometry wrong, you'll end up with a useless service no matter how attractive its technology is.

What's more, the most basic features that determine whether transit can serve us well are not technology distinctions. Speed and reliability, for example, are mostly about what can get in the way of a transit service. Both buses and rail vehicles can be fast and reliable if they have an exclusive lane or track. Both can also be slow and unreliable if you put them in a congested lane with other traffic. Technology choice, by itself, rarely guarantees a successful service, and many of the most crucial choices are not about technology at all.

TRANSIT AND URBAN FORM

Many of the “plumber’s questions” about transit will be easier or harder because of the way your city is laid out. The physical design of cities determines transit outcomes far more than transit planning does. Your particular location in the city, and the nature of the development and street patterns, will govern the quality of transit you can expect. For that reason, one of the most urgent needs related to transit is to help people make smarter decisions about where to locate their homes and businesses, depending on the level of transit mobility that matters to them.

These decisions, aggregated across the whole population and over many years, can change the shape of your city for better or for worse. Ultimately, our cities grow and change due to individual decisions about where to locate things. It may seem that the developers, planners, and politicians are making these decisions, but all citizens are part of the market that they serve. So everyone contributes to determining whether transit can work in a particular city or development. Chapter 14 is about how to make those choices more wisely.

WHO AM I? WHO ARE YOU?

Finally, because so little agreement exists about what constitutes expertise in transit, it is only fair that I answer the question: “Who is this guy, and why should I care what he thinks?”

Since 1990, I have been a consultant specializing in transit service design—that is, designing transit networks and their schedules to provide the best possible service to a community, according to the values each community expresses. I’ve also worked as an in-house consultant in major transit agencies, serving temporarily as part of their staffs. In many cities, from small towns to large urban areas, I can point to places where my service designs are working on the street and have made a clear difference in the relevance of transit. The core pleasure of my professional life is to see transit working well in the real world.

Ever since I was ten years old and began riding the city buses across Portland, Oregon, to school, I have been an attentive customer of transit, constantly musing about how this or that could work better. Phoning the planners at Portland’s transit agency, TriMet, I found them interested in talking with me and receptive to new ideas, so I never learned the debilitating cynicism that so many citizens feel—the sense that their transit agencies are so stuck in their ways that there’s no point in providing input. Though I sometimes criticize them, I am proud of Portland’s city and regional governments, and continue to admire what they are achieving.

Finally, I spent my twenties doing a PhD in a literature field, so I’m extremely sensitive to the workings of language. Throughout the book, I’ll point out situations where word choices matter and where the structure of our language may be preventing us from thinking clearly. But enough about me. Who are you?

In writing this book, I imagine that you, the reader, are a curious and thoughtful person who cares about whether we find our way to more rational forms of urban mobility. Perhaps you work or study in a field related to transit, such as land use planning, traffic engineering, or real estate development. You may be a community or business leader, a journalist, or a committed activist. Perhaps you have come to transit issues from another concern, such as the economic development of your city, or the rights of the disabled, or the well-being of seniors, or some form of environmental-

ism. Perhaps you have used and admired the transit systems of other cities and wonder why those systems aren't possible where you live. Perhaps you are simply a citizen who doesn't like what is happening to your city, who is not sure how to evaluate the transit disputes flaring around you, and who wants to hold more confident opinions. This book is for you if you come to the question of transit with clear ideas about what you want for your city and curiosity about how transit could help you achieve that.

WHERE ARE YOU?

The core ideas of this book apply anywhere in the world, but my focus is on the United States, Canada, Australia, and New Zealand. These countries share not only affluence by world standards but also an important feeling of spaciousness that comes from having a lot of land and only a few centuries of urban history. For that reason, these countries have led the world in building low-density, car-dependent cities—the hardest and most interesting challenge for public transit. Still, readers in other developed countries will find most of the book useful as well.

As for the less affluent or “developing” world, the geometry is the same but one crucial aspect is different. The cost of running transit in wealthy countries is usually dominated by the cost of labor—drivers, mechanics, and so on. In the developing world, labor is much cheaper, and fewer people own cars, so transit is cheaper to operate and has a larger customer base. That means transit is often a profitable small business—as small as one driver and his bus—and, therefore, is often abundant but poorly organized. Cheaper labor also means that smaller vehicles are economical, so developing world cities tend to have a diverse range of minibuses, shared taxis, and the like.

Although transit's costs and markets differ in the developing world, the same limitations of urban space do apply. Often, the abundance of minibuses and shared taxis is enough to create near gridlock even without many private cars. Today, the developing world is the site of frenetic development of modern bus systems, bus rapid transit, and sometimes rail transit. As these countries grow more prosperous, they face many of the same issues that we in the wealthier countries face now, so I hope readers in the developing world will find the book useful as well.

WHO ARE “WE”? WHO DECIDES?

Throughout this book, I use “we” to mean you and your fellow citizens in a democratically governed city, along with me as your hired transit planning expert. I assume, in short, that as a citizen, you’re someone whose opinion matters, and that if you get together with your fellow citizens and think together with the benefit of an expert’s advice, you can arrive at your own resilient view of what you want transit to do—and thus of what kind of transit you need.

If you’ve spent any time following transit politics in a big urban area, this notion may seem naive. Perhaps you’ve dealt with cities where a tangle of government agencies works on transit, often quarrelling more than they cooperate. Perhaps you’ve dealt with bureaucracies that seem defensive, trapped by their own fears and habits, and unwilling to engage you in respectful conversation.

Perhaps, too, you suspect that a small, self-interested elite really makes all the decisions that matter, and that public consultation is just a show. If so, you’re probably wrong about that. In twenty years in this business, I’ve seen plenty of frustrating, confused, and hijacked decision processes. If I wanted to be bitter and cynical, I would have the experience to justify it. But I’ve seen another pattern: the involvement of more informed and caring citizens—expressing themselves with courtesy, clarity, and persistence—almost always leads to better outcomes.

In any debate, there will be people at the decision table who care only about their personal interests and needs. If those people have the decision table to themselves, you’ll get decisions that serve those interests but may not serve your city.

But in situations where lots of citizens care, and choose to learn a bit about transit so that they can advocate more clearly and confidently, better decisions get made, decisions that lead to better mobility, a stronger economy, a more just society, or whatever goal the community is pursuing. *Political leaders make good decisions when informed and caring citizens want them to.* It’s only when they sense that citizens have given up or don’t care that they may let narrower interests carry the day. I see this pattern over and over.

So if you’re willing to learn a bit about how transit works, what it does well, what it doesn’t do, and how it fits into the larger challenge of the city, your opinion will count.

RESPECT FOR TRANSIT

Throughout this book, you'll be asked to develop a respect for transit not just as a tool but also as an area of expertise. Many people look at transit lines running back and forth in their city and feel that it *looks easy*. Many other people think they understand transit because they understand something that's connected to it, such as urban design, operations management, economics, or traffic engineering. Get all those people in a room and they start arguing and talking past one another, often hurling jargon as a weapon. Other citizens listen to a bit of this and decide that transit is impossible to figure out.

It's actually not hard to understand transit and how it can serve our values and needs, if we approach it with some respect. Let's try.

Continue the Conversation

As you read this book, you'll have questions. You are welcome to submit these at <http://www.humantransit.org>, using the email button under my photo. I will post replies to frequently asked questions, including links to further sources, at <http://www.humantransit.org/faq>. Please join me in continuing the conversation.

WHAT TRANSIT IS AND DOES

There are several ways to define public transit,^a so it is important to clarify how I'll be using the term. Public transit *consists of regularly scheduled vehicle trips, open to all paying passengers, with the capacity to carry multiple passengers whose trips may have different origins, destinations, and purposes.*

Let's take this definition apart:

- “*regularly scheduled vehicle trips*”: Transit is provided by a vehicle running on a regular schedule or pattern. There is room for variation in routes and schedules. Demand-responsive services, for example, may vary their routing according to customer requests, within set limits. But at its core, transit service must be predictable so that different people can plan around it without coordinating directly with one another. This feature is the crucial difference between transit and other ways of sharing a ride.
- “*open to all paying passengers*”: The word *public* in *public transit* means “open to the entire public.” This word can be confusing in debates about whether transit should be operated by the government or by the private sector. In the developed world, where wage costs are high, transit is usually subsidized by government, but it may still be operated either by government or by private companies. In those conversations, *public transit* can be misunderstood as meaning “transit operated by the public sector—that is, government—rather than by private

^a In Great Britain and most other Commonwealth countries outside North America, the term is not “public transit” but “public transport.” My definition would be the same.

companies.” That is *not* the meaning in this book or the prevalent meaning in the developed world. Even privately operated transport services are expected to welcome all paying customers; in fact, the failure to do so can become a civil rights issue.^b

- “*that can carry multiple passengers*”: The ability to carry many people with a single vehicle is the defining virtue of transit, and the most basic measure of its efficiency.
- “*whose trips may have different origins, destinations, and purposes*”: Transit, in the sense used in this book, does not include:
 - carpools and vanpools, where several people with the same destination share a ride;
 - school buses, where school is the only origin or destination served;
 - a family in their minivan, or any other group that’s intentionally traveling together;
 - taxis, which carry a small number of riders at the time, typically all with the same origin and destination.^c

There are many forms of multi-occupant vehicles, all of which are better for the environment than the same individuals each driving alone. Carpools, school buses, and shared taxis are all useful parts of a city’s transportation mix, and sometimes demand can be shifted between these services and the formal public transit system. But they are not *public transit* as this book, and most of the industry, uses the term.

At its core, transit is about multiple people riding in one vehicle even though they are not intentionally traveling together or even going to the same places. The core challenge of transit design, then, is how to run vehicles so that people with different origins, destinations, and purposes can

^b A service open only to a subset of the public—such as paratransit vans that serve disabled persons on demand—can count as public transit if it specifically serves persons who cannot access regular transit services. The spirit of paratransit is to serve people who cannot use regular transit service for reasons of disability, so even though it’s closed to other riders, it still serves the goal of a total service offering that is available to everyone. The US Americans with Disabilities Act, for example, requires paratransit service to exist only when and where fixed route services for the general public exist.

^c Taxis are sometimes used as transit vehicles at low-demand times, and in these cases they satisfy the definition of transit that I’m using here. However, taxis hired for the *exclusive* use of one person or party—the usual taxi model in the developed world—are outside of this definition of transit.

make their trip at the same time and will be motivated to choose transit to do so. This book is all about that challenge.

TRANSIT'S ROLE IN A COLLABORATION OF MODES

While this book is about transit, I never imply that transit is or should be the dominant alternative to the private car. Many ways of sharing vehicles have important roles to play in the larger project of reducing car dependence. These include many forms of carpooling and vanpooling, which typically carry people from a similar area of origin to a common destination, as well as “carshare” programs, which provide members with hourly self-service car rental, thus reducing a household’s need to own cars. These programs, commonly supported by transit agencies,^d are important complements to transit, though they are not this book’s focus.

In focusing on transit, I am also not denying the role of the “active modes,” such as cycling and walking. Quite the opposite. Virtually every transit rider is also a pedestrian, so transit ridership depends heavily on the quality of the pedestrian environment where transit stops.^e The ability and willingness of people to walk a short distance to a stop or station is what makes it possible to gather many people with many intentions on a single vehicle, which is the essence of transit’s project.

Cycling, meanwhile, is growing rapidly in many New World cities, at least those that have made some effort to accommodate it. But even in the most bicycle-dominated countries, such as the Netherlands, transit has a crucial market. Local bus service has a somewhat smaller role there because bicycles take so much of the short-distance market, but in the longer-distance market, trips over 3 miles (5 km)^f or so, cycling and transit reinforce each

^d “Agency” is a common North American word for the organization running public transit, so I’ll use it in this book. Your city’s term may differ, as there are many ways public transit can be organized within government.

^e Throughout this book, I will have to ask some disabled persons and their advocates to forgive my use of the terms *walking* and *pedestrian*. These terms are sometimes considered objectionable because a person in a wheelchair is technically not walking or traveling by foot. My use of these terms explicitly *includes* persons in wheelchairs and similar devices that allow them to travel at typical pedestrian speeds.

^f Throughout this book, “km” preceded by a number means kilometers, while “m” means meters.

other. Longer-distance “rapid transit” services (rail and bus) run fast by not stopping often, but their stations feature masses of bicycle parking. The bicycle becomes an ideal tool for extending the reach of a rapid transit station, reducing (but not eliminating) the need for slower local bus and streetcar services.

Many cities and transit agencies are looking at how to expand the potential for these “cycle + transit” trips. These efforts include enhancing bicycle storage opportunities at stations as well as allowing cyclists to bring their bikes onboard, at least during low-demand times when there’s room to spare. These strategies have the potential to build the market for both cycling and public transit.

So transit has the potential for a mutually beneficial relationship with most of the other alternatives to the private car.

- *Walking* is an intrinsic feature of almost all transit trips, so all transit advocates must be pedestrian advocates. Transit outcomes depend heavily on the nature of the walking required, including both how long the walk is and how pleasant it is.
- *Cycling* can compete with local transit but tends to complement longer-distance rapid transit, especially when investments are made in secure bicycle storage at stations.
- *Carpooling* is a crucial tool for regularly scheduled commutes, especially to lower-density employment centers, such as business and industrial parks, that are not dense enough to attract high-quality transit.
- *Carsharing*, a form of short-term car rental, is essential in cities that want to encourage lower levels of car ownership, at least in their denser neighborhoods where the space requirements of private cars are hardest to meet. Carsharing eliminates the temptation to own a car that you only need once or twice a week, by providing the cheaper option of shared cars for these purposes.

So even as these other sustainable transport modes grow, we will need public transit. Among the sustainable transportation alternatives, public transit is unique in two crucial respects. First, only public transit can carry large numbers of people in a single vehicle with a single driver,⁸ even as

⁸ Most public transit vehicles have one on-board employee, whom I’ll colloquially call the driver. Many US transit agencies prefer the word *operator*, but that word would sound

these people travel from different origins to different destinations for different purposes. At the intense levels of demand found in high-density cities, public transit is an efficient use of both energy and scarce urban space and is often the most attractive option for trips that are too long to walk or cycle.

Second, public transit delivers people from one part of the city to another *as pedestrians*, eliminating all the challenges of storing a personal vehicle. The pedestrian is the foundation of contemporary urban design, because walking is the only form of transportation that doesn't feel like transportation at all. Walking is also an ideal mode for both health and sustainability. If you want to encourage pedestrian life, you need to connect pedestrian-intensive places to one another in a way that the pedestrian can use. Transit can be ideal for this purpose.

FIXED OR FLEXIBLE?

To make a vehicle trip useful to many people who are not coordinating with one another, the vehicle trip has to be predictable. That's why, in the developed world, transit is dominated by *fixed* services; on these, transit vehicles follow the same path, at the same time, day after day, so that customers can plan around the pattern. Fixed services are the most efficient form of transit in terms of the ability to carry many passengers for each hour of the driver's time, so they have come to represent well over 99 percent of transit ridership in the United States.

The rest, accounting for less than 1 percent of ridership, are various kinds of flexible or demand-responsive services, where the routing followed by a transit bus or van can change based on customer requests. Although flexible route services are an area of great innovation, they remain limited because they're intrinsically less efficient. Taking a different route depending on customer requests, as flexible routes do, takes more of the driver's time for each passenger's needs. So flexible routes tend to be useful where the overall demand is low, or for specific populations whose needs aren't met by fixed services, such as some disabled persons. I'll return to

too vague in a book for the general reader. Note, however, that a driver's job varies considerably from one service to another, and in some cases, such as certain rapid transit trains, they aren't really driving at all.

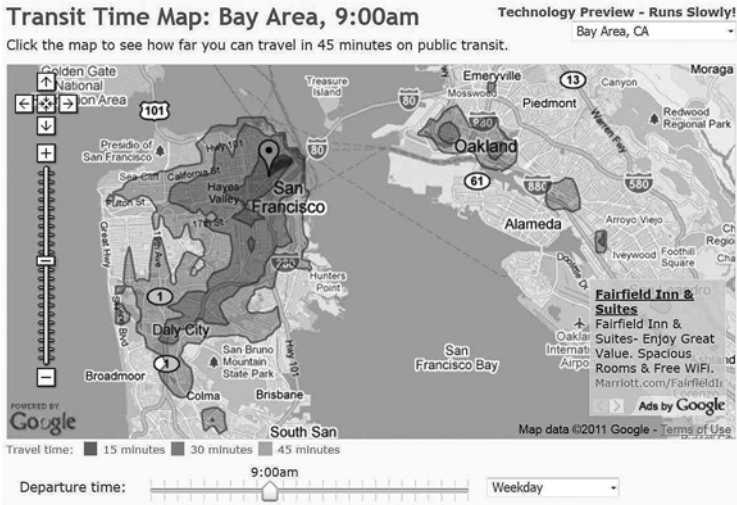


Figure 1-1 Walkscore.com online travel time map showing the area reachable by transit from a user-specified location in a user-specified period of time. Bands represent 15-minute intervals. *Source:* Walkscore.com

this issue in chapter 10, but for now let's focus on the services that the other 99 percent ride: the fixed services.

PERSONAL MOBILITY: THE FREEDOM TO MOVE

In 2009, we began to see web-based tools that allow you to enter an address and see where you can get to from there on transit in a fixed amount of time. Figure 1-1, for example, is the output from WalkScore.com when queried by someone near the San Francisco Civic Center at 9:00 am.¹

These tools aren't for planning a trip; rather, they're for visualizing your freedom. That freedom is transit's product. The core product that arises from transit is personal mobility, by which I mean *the freedom to move beyond your walking range*. Mobility is a controversial concept, so it's worth taking a moment to clarify how I'll use the term and why. Readers who are less interested in theory may skip the rest of this chapter, but anyone who thinks theoretically about transportation or urbanism will find this section crucial.

More on Personal Mobility

For more thorough discussion of the concept of personal mobility and how it relates to prevailing concepts of mobility and access in urban planning theory, see <http://www.humantransit.org/01box.html>.

A quick survey of definitions of mobility turns up a range of slightly different ones, indicating some ambiguity in the word. Here are a few:

The condition of moving freely.²

Ease of moving about.³

Movement of people or goods within the transportation system.⁴

By defining the term *personal mobility* as a freedom, I mean something close to “ease of moving about,” but this is not at all the same as *movement*. The first describes a degree of freedom, while the second is a result of some people choosing to exercise that freedom.

If you think of *person movement* as transit’s product, as many conventional measures do, the output of transit that matters is passenger-miles or passenger-kilometers. A passenger-mile is one passenger carried for 1 mile. (Fifty passenger miles, for example, could mean one passenger carried for 50 miles, or 50 passengers each carried for 1 mile.) As an approach to measuring transit’s output, this concept of quantities of movement is troublesome. It doesn’t measure how readily people got to where they were going; it just measures how far they were moved. Most of the time, though, our travel isn’t motivated by a sheer desire for movement; it’s motivated by the need to do something—to make some kind of economic or personal contact—that is too far away to walk to. In most cases, we don’t want movement. We want access.

Access is the ability to complete some desired personal or economic transaction. Your mobility can be visualized as where you can go in a given time. Access is how many useful or valuable things you can do. If a new

grocery store opens near your house, that addition doesn't improve your mobility but it does improve your access. You can now get your groceries closer to home, so you don't need as much mobility as you did before. You can also improve your access (but not your mobility) by working at home instead of commuting, downloading music instead of going to a store, or moving in with your romantic partner. In other words, much of the work of access is about eliminating the need to move your body around the city in order to complete the economic and personal transactions that make up a life.

As Todd Litman⁵ points out, access can be achieved in three ways. You can (1) travel to the thing you need, in which case you need mobility; you can (2) obtain it via telecommunications; or you can (3) relocate either yourself or the desired thing so that both are closer together. Urban redevelopment, which adds new destinations close to you, supports option 3 at a large scale.

Transit plays a role in both option 1 and option 3. Transit provides mobility directly, but it can also influence urban development, which in turn can improve access. For example, if a new subway station opens near your house, you get improved mobility. But if the station leads to new development around it, you may get a new grocery store close to home. Even if you never ride transit, that's an access improvement for you.

Still, transit's role in mobility is direct, whereas its influence on redevelopment is indirect. Transit may lead to access-improving development, but only via several intermediate and unreliable steps. You can build a transit line and still not get new development if any of several other things don't fall into place—including zoning, economic growth, cooperative neighbors, courageous developers, and bankers willing to lend. In that case, the new transit project doesn't improve access at all, unless it has improved the first kind of access: mobility.

So personal mobility—the freedom to move—is the direct product of public transit. Mobility doesn't always generate movement, but it does generate happiness. For this reason, people will resist locating in places where it seems to be denied.

If we want cities to be built in ways that require less travel—that is, if we want cities with better access—we will need to ensure that those cities still have generous transit mobility. We need to show that if you locate in a

transit-intensive place, you will be able to get to lots of places that matter to you, on transit.

So in a book on transit, I'm going to insist that personal mobility—the freedom to move—is still transit's primary product. Again, mobility is only one dimension of access. The other two are urban redevelopment and telecommunications, both of which can reduce the need for travel. But mobility is the kind of access that most people expect transit, in particular, to deliver.

Chapter 14 will look more at urban form and all the ways we can change it, but we can't use transit to create better cities unless we first understand how transit does its primary task of providing personal mobility. Transit must focus not just on city-building impacts but also on the perspective of someone who needs to go somewhere, and get there soon, to address an immediate need. This person isn't thinking about how better transit might help transform the city, but rather, "I need to be there!" We must figure out whether transit can help, and if so, how.

2

WHAT MAKES TRANSIT USEFUL? SEVEN DEMANDS AND HOW TRANSIT SERVES THEM

If you spend any time inside the offices of a transit agency, you get used to seeing messages like “The customer comes first!” and “Service is our business!” Posted in the elevator or in the lunchroom, these messages are supposed to focus employees on a particular mission called “service.”

But what kind of service do we need to provide so that people will use it? What is this mysterious thing called “service” anyway?

The most common answer is “Ask the customer!” As in any business, transit customers have needs, desires, and dislikes that must be a starting point in designing and operating a transit system. Most transit agencies do listen to public comments and demands and sometimes change direction because of them.

But most public comments are narrow and self-interested. A person wants transit to stop in a particular place, or not to, because of their personal needs or desires, not because it will help make the system more useful to the entire community.

To plan service for a whole community, including the vast majority who never comment, effective network planners look beyond self-interested demands and think more broadly about what motivates people to use transit. This doesn’t mean substituting our judgment for the customer’s, but it does mean trying to discern underlying patterns in the diverse comments

that agencies receive, and thinking about how various service changes would improve transit's ability to attract riders and fit the larger goals of the city or region it serves.

OUR SEVEN DEMANDS

In the hundreds of hours I've spent listening to people talk about their transit needs, I've heard seven broad expectations that potential riders have of a transit service that they would consider riding:

1. It takes me *where* I want to go.
2. It takes me *when* I want to go.
3. It is a good use of my *time*.
4. It is a good use of my *money*.
5. It *respects* me in the level of safety, comfort, and amenity it provides.
6. I can *trust* it.
7. It gives me *freedom* to change my plans.

I've listed the demands in the order in which you, as a customer, usually evaluate them. Generally, you would first evaluate transit in terms of whether it exists at all in the place where you need it (demand 1). Then you would consult a schedule and determine whether it's there *when* you need it (demand 2). Next, you might compare the cost of transit (in money and time) with the benefits (demands 3 and 4) to decide whether transit is worth trying compared to your alternatives.

Now you are ready to try the service. You notice whether you feel comfortable and respected as a passenger, and whether you can put your travel time to good use (demand 5). If you become a regular customer, you start noticing whether the service works the same way day after day—in short, you decide whether you can trust it (demand 6).

Finally, as your own needs vary from day to day, you begin to discover how well the service responds to those changes (demand 7). Can you get home, or to school, in the middle of the day to tend to a sick child? Can you, on the spur of the moment, stop off at a cinema and see a movie, knowing that it is still possible to get home 3 hours later? Can you quickly figure out how to make other trips elsewhere in the city, to visit friends

across town, or to attend medical appointments, or to go shopping? In short, does the transit system help you to feel free—able to enjoy all the riches of your city in a spontaneous way?

These seven demands, then, are dimensions of the mobility that transit provides. They don't yet tell us how good we need the service to be, but they will help us identify the kinds of goodness we need to care about. In short, we can use these as a starting point for defining *useful* service.

Side Issues and Side Effects

Do these demands encompass everything that people ask of their transit agencies? Of course not. There are two other important categories of demands.

First, a transit agency hears feedback about how the agency functions as an employer, as a company, and as a corporate citizen. Such issues include fairness to workers, discrimination, sponsorship of community events, donations to charities, relations with other governments, relations with the media, and so forth. These are all important, but they are common to all agencies and companies within a community, regardless of their product, so focusing on these will distract us from understanding public transit in particular. For that reason, I will set these aside for this book's purpose. You don't need to understand transit to understand these issues.

Second, agencies get many comments about the side effects of their equipment and facilities, and these considerations are enormous factors in decisions about new transit infrastructure. The big side effects of transit are emissions, noise, vibration, and the way various transit vehicles affect the look and feel of the urban environment. Many of these are huge issues, and a great deal of money is spent mitigating these impacts. But these comments aren't about the basic work of public transit, either. An especially noisy and polluting bus has the same impact regardless of whether it's a public transit bus or a tour bus; in fact, the same bus may serve either purpose. Other technologies attract advocacy because of positive side effects. Streetcars, for example, are popular in part because of how they look and feel in the urban streetscape.

In considering side effects, we need to think about transit the way we think about the fire department. Firefighting has lots of side effects, including the space the big trucks require and the noise of their sirens. Now

and then, those effects become so objectionable that people complain about them, and fire departments have to respond to those complaints, as they should. So there's a values trade-off here, an example of the plumber's question: Should we reduce the effectiveness of firefighting a little in order to mitigate some of its impacts?

Still, all sides of debates around firefighting impacts can see that there is a job called "firefighting" that is the defining purpose of these companies. That doesn't mean that it's the only thing that matters or that its efficiency should be ranked above all of its other impacts. But discussions about impacts usually reflect a shared understanding that the firefighters' main job is to put out fires and that everyone has an interest in seeing that job done well.

When it comes to transit, the balance of power in these conversations is different. Not everyone recognizes that transit has a defining activity—personal mobility that does not require personal vehicles—and that a transit agency has to maintain a primary focus on that even as it tries to manage its side effects. Side effects are important, but they should not create confusion about the defining mission of transit, which is to provide personal mobility. The seven demands for useful service are all about that mission.

THE ELEMENTS OF USEFUL SERVICE

So how does transit meet the seven demands that we identified above? How do we translate the demands into specific things that transit agencies have to provide and measure? Figure 2-1 shows how each of the seven demands relates to the key measurable features of a transit service. Don't worry if it looks complicated. We'll step through the key points. Still, you may find it useful for reference as we explore how the various transit concepts are connected.

Demand 1: "It Takes Me *Where* I Want to Go"

The first demand—"it takes me *where* I want to go"—involves two key measurable features of a transit system:

First, the location of *stops and stations* determines how close transit service comes to each place that anyone might want to come from or go to.

The Seven Demands of Useful Service

How Transit Serves Them

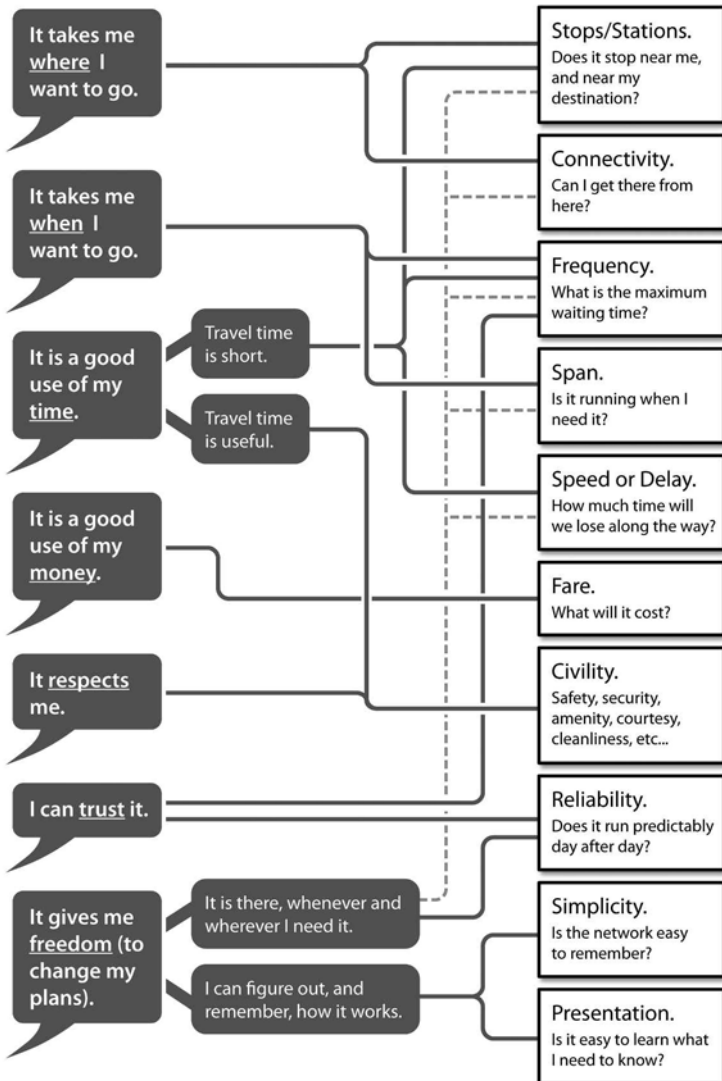


Figure 2-1 Seven desires for useful transit, and how transit serves them. *Credit:* Eric Orozco

We'll discuss stops and stations in chapter 5, and the lines that they form in chapter 4.

Second, *connectivity* is a measure of whether transit links the place you want to go (your *destination*) with the place you are now (your *origin*). Stops near your origin and destination do not guarantee connectivity: a transit agency may serve both point A and point B but be unable to take you from A to B by a reasonably direct path. Chapters 12 and 13 explore connectivity.

Demand 2: “It Takes Me *When I Want To Go*”

The second demand also encompasses two measurable features.

First is the question of whether transit runs at all when you need it; the answer to this is the *span* of service. Span is indicated by the scheduled time of the first and last trip in each direction.

Next is the question of whether the service runs often enough that you can leave when you really want to go. The measure of how often transit runs is *frequency*. Frequency is usually described by the number of minutes between consecutive trips, so—as in golf—lower scores are better. A service that runs every 15 minutes, for example, is twice as useful, and twice as expensive to operate, as one that runs every 30 minutes.

This number—the gap between consecutive trips in minutes—is called the *headway*. If we change a frequency from every 30 minutes to every 15 minutes, we say that we've doubled the frequency by halving the headway. Again, when you hear the term *high frequency*, that means a *low* headway, such as service every 10 minutes or less. A high headway, such as 60 minutes, is a low frequency.

Frequency is one of the most misunderstood concepts in transit. We'll return to it often throughout the book, especially in chapter 7.

Demand 3: “It Is a Good Use of My *Time*”

Demand 3 includes all the ways of making travel time useful to the passenger. These efforts can involve providing reading lights, electrical outlets, phones, Internet access, and other facilities that enable a passenger to work, sleep, or do something else valuable while waiting and riding.

Still, we want travel time to be short. Travel time is one of the most critical elements of a passenger's decision about whether to use transit rou-

tinely, and for this reason, the models that predict ridership give it huge weight.

When we think of saving time, we usually think of speed. When we're driving, the average speed that we'll achieve is the measure of how fast we'll cover the distance to our desired destination. When people who usually by travel by car think about a transit issue, they often focus on how fast the vehicles can physically go.

But for transit, the vehicle speed is a small part of the picture. What matters is how much time it takes to complete an *entire* desired trip. So to understand this demand, we have to look in more detail at all the phases of a trip and the potential for delay involved with each. We'll come back to that at the end of this chapter, and we'll dissect speed further in chapter 8.

Demand 4: "It Is a Good Use of My Money"

This fourth demand is relatively simple, since *fares* are the primary monetary cost of transit use. People compare the cost of using transit with the cost of getting somewhere in some other way, and this can strongly influence their decision. Chapter 11 explores fares.

Demand 5: "It Respects Me"

At its core, the fifth demand is about whether the rider feels valued as a customer, as a citizen, and as a human being. Obviously, this demand is so subjective that it could come apart into a million values.

Fortunately, most of us do agree on some things that contribute to our sense of being respected. For example, comfort, courtesy, safety, and security all signify we are valued as human beings and as customers. Some of these features also help to give value to our time, so they follow also from the third demand.

All of these values fall under a large category that I'll call *civility*. These values are not about transit's ability to transport us but, rather, about its ability to treat us like human beings while it is doing so. By "civility," I mean widely shared values unrelated to travel time or cost—such as comfort, cleanliness, courtesy, safety, security, and amenities that give value to our time.

Although they can't all be quantified, the elements of civility in transit are, by definition, widely agreed upon within a culture. Everyone understands the importance of comfort, cleanliness, courtesy, safety, and security. Some level of civility is essential if we are to carry a diverse range of customers. When a transit system gets the reputation of being “only for poor people,” despite good speed and frequency, it is often because its civility has fallen so low that most people don't consider using it as an option. Often, by this time, other important values will have declined as well, because the only people who care about them are politically powerless.

Definitions of civility may vary from one culture to the next. In some cultures, for example, civility may include separate seating areas for men and women. The acceptable standard for certain elements of civility may also vary by culture. Passenger railways in India may not offer the personal space and comfort that a North American visitor would expect, but they do meet the expectations that prevail in India, so they're popular. Civility includes, by definition, whatever expectations are widely shared within the culture.

Civility is hugely important, and problems in this area often send signals that a transit system is uninterested in serving the whole community. Still, the work of creating civility in transit is not much different from what it is in other endeavors. The customer service skills of a merchant, for example, are easily transferred to the customer-contact role of a transit information agent or, for that matter, a bus driver. Creating a civilized waiting environment for transit is not that different from designing waiting areas for other businesses and government functions. So, this is an area where we can rely on expertise from other fields, and indeed much good work is being done on the topic. For that reason, in a book that focuses on transit in particular, I won't discuss civility in great detail.

Demand 6: “I Can Trust It”

Your transit service may get you to work on time 90 percent of the time, but if you're going to a meeting that's crucial for your career, you might not want to risk the one-in-ten chance that it will let you down. Our ability to trust a transit operation is called *reliability*, though frequency is also an important aspect of trust because it reduces our dependence on the reliability of any single transit vehicle.

Demand 7: “It Gives Me *Freedom* (to Change My Plans)”

Freedom is a crucial sensation, and in most places it is the private car’s crowning virtue. When limited transit schedules interfere with people’s lives—forcing them, for example, to decline a last-minute dinner invitation or cut short the family’s day at the zoo—we see why transit is not the mode of choice for more of the trips we make.

In transit, the real test of freedom is spontaneity. Can I change my plans suddenly? Can I get home if I need to, or to my child’s school if something comes up? Can I simply move freely around my city, following whatever impulse I may feel at the moment? Some transit systems approach that level of convenience, at least in dense cities. In some of those same cities, you’ll find that your car is an encumbrance. If you’ve ever been stuck in traffic on a busy downtown street while pedestrians and cyclists flow past you and subway trains zoom beneath you, you know that sometimes your car can become your prison.

If anything, the demand for spontaneity is increasing as families diversify. In the United States, for example, most households no longer have a parent devoted solely to the home; the norm is increasingly a couple who both work or a single working parent. These people have complex lives in which “something may come up” during the day that requires unexpected travel. Corporate styles are demanding more self-management and accountability from their employees, which often means that employees may go to work in the morning without knowing exactly when they’ll come home.

This demand for spontaneity is met mostly by values that we’ve already listed. A transit system offers freedom if it offers *frequency* and *span* (so that there is service whenever you suddenly need it) and a reasonable average *speed* compared to your alternatives. We might think of these values as “squared” by their role in meeting the demand for freedom in addition to their role in meeting one of our other basic demands.

But freedom is also the biggest payoff of *legibility*. Only if you can remember the layout of your transit system and how to navigate it can you use transit to move spontaneously around your city. Legibility has two parts: (1) simplicity in the design of the network, so that it’s easy to explain and remember, and (2) the clarity of the presentation in all the various media.

No amount of brilliant presentation can compensate for an overly complicated network. Anyone who has looked at a confusing tangle of routes on a system map and decided to take their car can attest to how complexity can undermine ridership. Good network planning tries to create the simplest possible network. Where complexity is unavoidable, other legibility tools help customers to see through the complexity and to find patterns of useful service that may be hidden there. For example, chapter 7 explores the idea of Frequent Network maps, which enable you to see just the lines where service is coming soon, all day. These, it turns out, are not just a navigation tool, but also a land use planning tool.

Another important legibility technique is to minimize the difference between the transit network and the street network, the latter of which you can assume your customers already know. Suppose you're in San Francisco and see a bus whose overhead sign says "38: GEARY 48th Ave." If you live in San Francisco, you probably know where Geary Boulevard and 48th Avenue are. So just a glance at this sign tells you that there's a bus that runs out Geary as far as 48th. The sign gives everyone who sees it a full sense of what the service does and how it might be useful. Clear information is often the best marketing.

This legibility is not the result of just a well-phrased sign. It's also the result of planners deciding that it's best for transit to just run the length of Geary Boulevard, rather than branch off of it in complicated ways that could never be described or remembered so easily.

As for maps and schedules, these used to be the transit agency's job, but the role is shifting to the private sector. Many transit agencies are now releasing their route and schedule data in standard formats, so that anyone can design printed maps or online applications to present it. This is a welcome change. There's no reason you should rely on a transit agency for maps and schedules, any more than you would rely on your government's highway department for maps of the highway network. Presentation of real-time information, which shows actual locations of vehicles rather than just what's scheduled, is also migrating to the private sector, where entrepreneurs are competing to create mobile phone software that will present this information quickly and compellingly.

Legibility may seem like a minor problem compared to the others. For people who use transit only for the same trip every day, legibility is not a problem; you just learn how to make your routine trip and ignore all the

other complexities. But if you want transit to be a primary transport mode for your city, then you want passengers to use transit for many purposes, not just a regular trip, and in this case the legibility of the system becomes a critical value. Anyone who has spent 15 minutes on hold waiting for a transit system's information line (and who only called because the maps, schedules, and website were not helpful) knows that the lack of legibility really does add to travel time and crushes any sensation of freedom.

What It Costs

Journalists love to quote the eye-popping numbers for the construction cost of major transit lines, but *operating cost* is the eternal reality that governs most transit budgets. If you don't understand the essence of operating cost, you won't understand planning proposals.

In general, operating cost roughly doubles if you

- double the total length of the transit lines you operate, either by extending existing lines or by creating new ones;
- double frequency, for example, by cutting the headway from 30 to 15 minutes; or
- double the duration of service, for example, by expanding from 8 hours a day to 16 hours a day.

But one popular improvement saves operating cost: increasing speed. In general, if you were able to cut the travel time of a service in half—that is, double its average speed—your operating cost would drop by up to half. That's because most operating cost is labor, so it varies with time rather than distance. Run faster, and you use labor more efficiently.

For more on operating cost, see <http://www.humantransit.org/02box.html>. For now, remember: route distance, frequency, and span all *cost*, but speed *saves*.

THE SEVEN PHASES OF A TRIP

Another useful way to sort the elements of useful service is to think about how they relate to the phases of a transit trip. Any trip on transit has the following phases or steps, and in each one we seek both an efficient use of our time and money and a civilized experience.

1. *Understanding.* First, you must form a sufficient understanding of the service, frequency, and fare to know how to make the trip. This is the role of *legibility* as discussed earlier.
2. *Accessing (at the origin).* You then walk, drive, or cycle to the stop or station where you will board the service. Here, there are limits to what your transit agency can do, apart from locating the stop at a logical place, making it a civilized place to wait, and providing parking options for your car or bike, where appropriate. Other than this, you're at the mercy of the city's street network and development pattern.
3. *Waiting.* Waiting is everyone's least favorite phase of a trip. It's governed mostly by *frequency* and *reliability*, but of course the quality of the waiting environment has a big impact. Waiting is also being transformed in interesting ways by personal technology, most notably by real-time information that tells you how long you have to wait.
4. *Paying.* We think of paying as spending money, but paying can also cost time, especially if you are unfamiliar with the system. If the driver collects the fares, your act of paying takes the time of everyone already on-board. That's why higher-volume transit systems (rail, bus, or ferry) are shifting to forms of fare collection that require you to buy a ticket before you board (more on this in chapter 11).
5. *Riding.* The time spent on the transit vehicle is governed by average speed (or *delay*) and *reliability*. The quality of the time, which is also important, is governed by both the quality of the vehicle and your ability to make use of the time, often through personal technology. Chapter 8 explores these issues.
6. *Connecting.* If your trip requires a connection, you'll repeat steps 2 through 5 for that connection. Minimizing the hassle of these steps is the work of *connectivity*, as we'll explore in Chapter 12. For example, fare systems that offer free connections eliminate the cost and hassle of repeating step 4.

7. *Accessing (at the destination)*. Finally, you'll repeat step 2 in reverse, traveling by some means from the transit stop/station to your actual destination. Again, the location of *stops/stations* (chapter 5) is the main transit feature that governs this access, but the design of the local area, which is usually outside a transit agency's control, is also a major impact.

Notice again that frequency appears multiple times. Frequency rules step 3 (waiting) and counts again in step 6 if a connection is required. But frequency also plays a big role in step 1 (understanding). A very frequent transit service is one where you don't worry about a timetable, and that's a huge step toward making understanding easier.

How Much Does Each Phase Matter?

The phases of a trip raise an important question that's likely to have a very personal answer: Do you just want to minimize your total travel time? Or, do you dislike certain phases of the trip so much that you'd endure a longer trip in order to minimize them? (Some of us might also endure a longer trip in order to have a particular pleasure, such as a ride on a vehicle that they perceive as fun, but this is even harder to quantify, as the nature of fun is so subjective and variable.)

Many studies have looked at how different kinds of delay seem to influence people's decisions to use transit, mostly for the purposes of ridership prediction. There are many ways to frame this question, but the most useful for our purposes is: "How much does a minute of this time deter someone from riding, as compared to the deterrent effect of a minute of riding time?" Table 2-1 shows some figures cited in an influential US manual, derived from studies in eight North American cities dating from 1960 to 1995.¹

For example, when Table 2-1 says that walk time is weighted, on average, at 2.2 times riding time, we mean that 1 minute of walking time has as much deterrent effect as 2.2 minutes of riding time in determining whether someone will choose to use transit. For an extreme but simple example, if you could get to your destination in either 10 minutes of pure walking or 20 minutes of pure riding (with no wait or walk delay), these models say that, on average, people would rather have the 20-minute ride, even though they'd get to their destination 10 minutes later.

Table 2-1 Deterrent Effect of Various Kinds of Travel Time
(relative to ride time = 1)

Delay Type:	Walk	Initial Wait	Ride	Wait for Connection
<i>Governing Feature:</i>	<i>Stops, Stations (chapter 5)</i>	<i>Frequency (chapter 7)</i>	<i>Speed, Delay (chapter 8)</i>	<i>Frequency, Connections (chapters 7, 12)</i>
Minimum	0.8	0.8	1.0	1.1
Average	2.2	2.1	1.0	2.5
Maximum	4.4	5.1	1.0	4.4

Source: *Transit Capacity and Quality of Service Manual*, 2nd ed.

When you put it that simply, it's easy to say, "Wait, but I wouldn't do that!" Factors like these have to be used with care. They're the result of observing many people's travel behavior in complicated situations where many motivations overlap, so an observed behavior that appears to be a response to waiting time may actually be about something else, such as the quality of the waiting environment, the reliability of the service, or the availability of information. Analyses like the one discussed here are often cited as reasons why transit planners shouldn't expect customers to walk, or to make a connection, even if that makes their trip faster. But look at the minimums! All of these types of delay have, in various situations, been reduced to around 1.0. Achieving that would mean that for most people in most situations, all the elements of travel time are equally tolerable. That, in turn, would mean that with a manageable number of exceptions, the total door-to-door trip time, compared to the customer's alternatives, could be viewed as the main kind of time that matters to ridership.^a If that were

^a There will always be exceptions, but there can also be continuous efforts toward addressing those exceptions, efforts made simpler by a clearer goal. Right now, many people commuting long distances value a good on-board work environment, but wireless internet and smartphones will eventually improve the working environment on most long-distance transit services. Weather deters walking and waiting sometimes, but this can be addressed through weather-protected connection points, heated or cooled shelters, and the continuous awnings that some cities (such as rainy Wellington, New Zealand) require in business districts. If enabling people to get where they're going as

true, it would be much easier to decide, for example, that most people will walk farther to a faster service that will get them to their destination sooner. Suddenly, technical debates about transit would become exponentially simpler.

Is that possible? If it were our objective, we could certainly get much closer than we are now. Chapters 5 (on stop spacing), 7 (on frequency), and 12 (on connections) will look at these three kinds of travel time and how we can minimize their deterrent effect. Of course, we also have to care about in-vehicle travel time, which is the subject of chapter 8.

But there's another reason for optimism about bringing down the deterring effects of walking, waiting, and connecting, and that is today's revolution in information, which is not reflected in these twentieth-century figures. Most of these figures, for example, were developed in the absence of trip planning software, which allows you to evaluate your own transit options for a trip and decide for yourself if you prefer a longer trip that avoids walking or connecting. Without that knowledge, many passengers may have navigated by habit rather than choice because the available information encouraged that.

Real-time information is also transformative. Walking, waiting, and connecting are all made worse by uncertainty. Think about how different a walk to the bus or streetcar stop will feel if you know that the vehicle you want will be there in exactly 9 minutes. Even if you're in a hurry to reach your destination, you won't need to hurry while walking, so you'll enjoy the walk more. Waiting and connecting, too, become less onerous if you know the exact length of the wait, so that you can do other things with your time.

As mobile phone-based information sources become more sophisticated and universal, we will see a decline in uncertainty. Eventually, we may well reach a point where the length of the whole trip is really the only duration that matters. With clear, immediate, and reliable information, won't people find it easier, when they're in a hurry, to choose the fastest way?

fast as possible were the objective, a great many available innovations—in urban design, information systems, and network planning—could be brought to bear, and new innovation would have a clearer goal to pursue.